

PUNYA VENKATESH

AI Engineer | Multi-Agent Systems & RAG | IBM watsonx Orchestrate | Master of AI, Monash

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Work rights: Student visa (subclass 500) — 48 hrs/fortnight during semester, unrestricted during university breaks

PROFILE

AI engineer with more than 1 year of industry experience building **LLM-powered multi-agent systems** for enterprise clients, Currently Pursuing Master of Artificial Intelligence at Monash University.

Designed and delivered proof-of-concept agentic AI workflows on **IBM watsonx Orchestrate** for Agilent Technologies and Niagara Bottling, cutting incorrect prioritisation decisions by ~70% in benchmark testing against synthetic operational data.

Core stack: Python, LangChain, RAG pipelines, Hugging Face Transformers, SQL, FastAPI, Docker. Five IBM AI and security credentials.

PROFESSIONAL EXPERIENCE

Associate AI Engineer - 3K Technologies

Nov 2025 – May 2026

Bengaluru, India

- Designed and built a **6-agent, 7-tool Smart Prioritization proof-of-concept** on IBM watsonx Orchestrate for Agilent Technologies, automating task scheduling and ranking across service operations.
- Cut incorrect prioritisation decisions by **~70%** — measured against a baseline on synthetic operational data — by embedding structured data validation checks into the agent pipeline.
- Built enterprise post-quantum cryptography (PQC) readiness through hands-on delivery on IBM Guardium Quantum Safe Remediator and Cryptography Manager, developing working expertise in cryptographic discovery and remediation workflows.
- Enabled client-facing quantum-safe security demonstrations by preparing CBOM (Cryptographic Bill of Materials) artifacts and presentation materials alongside the IBM Quantum Safe Engineering team, translating PQC capability into business-ready use cases.

Data Science Intern - 3K Technologies

May 2025 – Oct 2025

Bengaluru, India

- Built a 4-agent, 5-tool agentic AI production dashboard proof-of-concept for Niagara Bottling with real-time monitoring, alert prioritisation and conversational, context-aware querying, cutting time-to-insight for plant managers by **~70%** in testing.
- Delivered role-based views tailored to each production team, demonstrating how operational visibility could be scoped per role.

Machine Learning Intern - Nerastech

Feb 2025 – Apr 2025

Bengaluru, India

- Built and fine-tuned optimised ML models for resource-constrained edge IoT devices, enabling predictive analysis and anomaly detection in limited-compute environments.
- Completed hands-on industrial training in data analytics, feature engineering and core machine learning algorithms, applying them to real business case studies under expert supervision.

PROJECTS

ResearchMind — Agentic AI Assistant for Scientific Discovery

- Built a multi-agent research assistant (LangChain, Groq LLaMA 3.3 70B, FAISS, Hugging Face embeddings) that pulls live papers from the arXiv API, answers questions with inline citations via a RAG pipeline, and runs a hypothesis agent to surface key findings, research gaps and novel hypotheses. Deployed on Streamlit.

CliniQ — Agentic Clinical Decision Support

- Built a 4-agent diagnostic system (LangChain, Groq LLaMA 3.3 70B, FAISS) that parses free-text patient symptoms, searches 36M+ PubMed papers live via the NCBI E-utilities API, returns ranked differential diagnoses with likelihood scores and cited evidence, flags emergency cases, and auto-generates downloadable SOAP notes.

LLM-Powered Document Summariser

- Designed an end-to-end RAG pipeline (LangChain, Hugging Face Transformers, FAISS) with optimised chunking and vector-based retrieval, enabling fast, accurate summarisation and querying of large documents. Deployed via an interactive Streamlit interface.

EDUCATION

Master of Artificial Intelligence — Monash University, Melbourne

2026 – 2028

- Coursework: Multi-Agent Systems, Deep Learning, Machine Learning, Predictive Analytics, Database Systems, Data Analytics, Data Visualisation.

Bachelor of Engineering (AI & Machine Learning) — RNS Institute of Technology, Bengaluru

2021 – 2025

- CGPA: 8.3 / 10

TECHNICAL SKILLS

AI / ML & GenAI: Agentic AI, Multi-Agent Orchestration, RAG, LangChain, Hugging Face Transformers, Prompt Engineering, Model Evaluation, Machine Learning

Programming & Data: Python, SQL, Java, C/C++, Pandas, NumPy, Statistical Analysis

Platforms & Tools: IBM watsonx Orchestrate, Docker, FastAPI, Streamlit, FAISS, Groq, Hugging Face Spaces, arXiv & PubMed APIs, Power BI, Excel, Exploratory Data Analysis

Professional: Stakeholder Communication, Cross-Functional Collaboration, Technical Presentations, Client-Facing Demonstrations

CERTIFICATIONS

IBM Generative AI for Code with watsonx Code Assistant — Technical Sales Intermediate

IBM watsonx Orchestrate — Sales Foundation

IBM Guardium Data Protection — Intermediate

IBM Guardium Data Protection — Sales Foundation

IBM Data and AI Foundational Course